

Typing and Winning

Testing a Simple Theory of Defensive Type Coverage and its Predictive
Power in Pokemon Battles and Redesign Champion Teams

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Abstract:

This paper tests whether a simple heuristic as defensive coverage can predict performance of Pokemon teams and a project to redesign the Champion teams of Kanto to Alola. The paper proposes Mean Weakness Exposure (MWE) as a new variable to consider for team compositions. The paper also asks whether there is a relationship to Stat Total across the team and win rate. Through a genetic algorithm to design test teams and using the variable, I redesigned the Champion teams to test my theory because I wanted to see if they could be improved, hopefully with MWE. After running their initial teams and new teams against the test data, the result showed no relationship to both MWE and Stat Total.

Introduction:

I got bored one day and thought "Is there a way to make Champion teams in Pokemon harder without just giving them legendaries?" The popular ROM Hacks are guilty of handing out Rayquaza to Steven and Arceus to Cynthia. I think it's a lazy design to just allocate a stat stick to them. I decided to try to design a way to build teams and see if they perform better than the original Champions' teams. I designed a variable of weakness exposure using the damage multipliers. The product of each type across the teams' exposures are averaged. I will call this Mean Weakness Exposure (MWE). My hypothesis is lower scores of MWE correlate with higher win rates. However, changing the teams drastically also affects the Stat Total of the teams. I decided to also factor in Stat Total changes and if they predict win rate as well. I decided on Champion teams because if this variable has any predictive power, the Champions will be the ideal characters to implement them on.

Rules:

What am I considering a Champion and which will be covered:

I am considering the last battle the player does before the credits roll as the Champion of that region. This includes Blue, Lance, Steven, Wallace, Cynthia, Ghetsis, Iris, Diantha, and Professor Kukui.

Team Composition Goal:

I want the team to be the best of what each region has to offer with each Champion's personal flair. Some of these rules will be met with disagreement but as a player I feel that they accommodate a reasonable level of restriction and freedom for them. If this theory works, the rules should not be an issue so long as MWE is different and it results in a different win rate. This is half a statistics paper and half a passion project.

Ace:

Each team uses their respective aces as a non-negotiable. Lance must have a Dragonite, Steven needs a Metagross, etc. There is a caveat about a “de facto ace” in the section about starters. They are also allowed to use the aces of other gym leaders or Elite Four members.

Legendaries and Mythics:

I love playing ROM Hacks. The only issue I get is when they design teams and give Cynthia an Arceus and Steven Stone Groudon, Kyogre, and Rayquaza. It feels cheap and I want these teams to not just be stat sticks. Post-game Pokemon like Rotom, Gliscor, Electivire, etc. are available for their teams. The non-Unovan Pokemon in the Black and White post-game are not included. I want the player to feel each Champion has intelligently designed their teams with the available Pokemon of their regions. Colloquially “Unovan Pokemon” don’t refer to the post-game, other region ones.

Starters:

Blue and Professor Kukui all have starters as their aces and I intend to keep that for their new teams. In HeartGold & SoulSilver Lance has a Charizard on his team regardless of the player’s choice of starter. If his team is optimal with a Charizard in it, then he can keep it. This is the only exception to this rule.

Items, Natures, Abilities EVs, IVs:

Items for the test data are ideal VGC picks for each Pokemon. The Champion teams initial and improved will have the same which is typically a Sitrus Berry for the Ace. Ghetsis and Iris have a Life Orb and a Focus Sash Diantha and Kukui have a Mega Stone and Z Crystal respectively and their improved teams will have that too. Natures, and Abilities will be decided by the contemporary VGC optimal builds used by each Pokemon selected into the teams. EVs are simple spreads of 2 main stats 252 and one with 4. IVs are max with the exclusion of Hidden Power Pokemon to get their ideal typing with minimal change. One IV stat corresponds to 1 stat at Lvl.100, I didn’t think these changes would have a meaningful impact.

Typing and Physical/Special Split:

Type interactions that were changed later will not be homogeneously applied. The region with its type chart will be applied. The Physical/Special Split will be applied in all regions excluding Kanto and Hoenn. The Steel changes will not be applied and no Fairy-type until Gen 6.

Methodology:

Method:

Originally, I wanted to develop as many combinations of teams as I could with the 48 most common Pokemon (from BLourneco’s survey) and design Champion teams to battle them and compare them to their original teams. Doesn’t sound that bad but if I wanted to play out a Swiss Tournament (teams vs, scores of wins and loses made, and pick the best performers), it would have taken 75.3 trillion battles (the human brain does not have an intuition for factorials). I decided instead to get a set of strong, varied teams and have the champions battle them to ease the CPU and waiting time. An initial population of 2048 teams were randomly selected and made to do 8 battles each. Through win rates, the best 12.5% would be selected for the next

generation. Included with the best performers were three new populations, a mutated one, a crossover inheritance, and a randomised group. Through 7 generations there would be a total of 32,640 battles and each team would battle 28 times. The populations halved from their predecessors until 8 remain. These 8 are the test teams.

After the first run of the Kanto dataset, I got hundreds of errors regarding the game being a tie after 1000 turns. I decided that after 200 turns the losing team by health percentage will forfeit to ensure winners. If the percentages are the same it will look at remaining Pokemon and total status effects. Luckily the AI is heuristic-based and not deep reinforcement. This meant it won't learn to anticipate the ending of a match and spam healing to win by default.

MWE:

The Mean Weakness Exposure is measured by each Pokemon's defensive resists and weaknesses, multiplied by each type and then averaged throughout the type chart. It is the average of resists by type across a team divided by total types.

$$\frac{1}{17} \sum_{i=1}^{17} 2^{ix}$$

(I think I did it right :))

The paper posits that lower MWE are good for the teams' win rates. The metric punishes specialisation and rewards diversity. Punishing with the weaknesses neglected and stacked up like Water types' and their Grass weakness stacks exponentially. Rewarding by pushing the type total towards zero each resist. Wailord, Whiscash, and Seismitoad's Grass score will be 32 ($2 \times 4 \times 4$). Goodra and Klingklang's Steel score would be $\frac{1}{2}$ ($\frac{1}{2} \times 1$). Immunity is scored high as $\frac{1}{8}$ due to the value of nullifying an enemies move completely.

Test data:

I designed each of the 48 most common (with some exceptions explained later) with Abilities, Items, EV and IV spreads, movesets, and Nature I and ChatGPT believe give them an edge. The data is supposed to represent the most optimised Pokemon that a challenger brings to the Pokemon League. The Generation breakdown is Kanto and Hoenn are Gen 3, Johto and Sinnoh Gen 4, both Unova sets Gen 5, Kalos Gen 6 X/Y, and Alola Gen 7. Abilities were their standard or the better one if there were an option between two e.g Study for Golem. Hidden Abilities were introduced in Gen 5 and used if legal and better for them (e.g Emboar's is Reckless but in Gen 5 it wasn't introduced thus not used). Items were legal for relevant Gens. Every Pokemon was given an item. The EV spreads were simple spreads (two 252 and one 4) for the two most relevant to the Pokemon. The IV spreads assumed perfect IVs unless they had Hidden Power. The move changes its typing based on the IV score. Some Pokemon have marginally reduced IVs to meet their ideal typing. Movesets were VGC expected combinations. Sweepers with damage and set-up, stallers with protection, recovery, and status, etc. Natures were typically sacrificing their non-dominant damage type for either Speed or their dominant damage.

All data was originally in Excel spreadsheets then converted to VS Code .txt files in the Showdown Import/Export format.

IQV:

Indices of Qualitative Variance (IQVs) of each generation represent the variance of Pokemon across their eight teams. This was to prevent the test teams having the same few Pokemon, thus being biased data. I want the Champion teams to improve their overall performance, not against a local maximum. Type variance within teams was not considered and was not too important as a lot of the Pokemon have large variation in their move types. However, this is not enough to ensure the champion teams aren't just beating a local maximum of teams of four Water-types. I sacrificed intrateam type variation for team strength. To hopefully counter this, the approach to Champion team selection was attempted blind to the test teams bar when I had to collect their IQVs to minimise bias. No IQV <0.96 was accepted.

Region	IQV:	$\sqrt{\sum f_i^2}$:
Kanto	0.974	106
Johto	0.960	140
Hoenn	0.969	118
Sinnoh	0.966	124
Unova 1	0.970	116
Unova 2	0.973	108
Kalos	0.975	104
Alola	0.971	114

Battle AI:

I found an article on how the AI works in Pokemon. There's a point system and not much that is worth mentioning to conclude it's not that great. A simplification of the AI was needed. I got ChatGPT to code me a competitive AI built on heuristics. I decided against a deep learning model and has cost the potential of the teams used to their best. However, the paper's purpose is MWE and Stat Total, not deep reinforced AI. It works with a score value determining what move is best as highest int value. The base damage move calculation is STAB by Effectiveness by Accuracy (STAB X Eff X Acc). An example is Musdale using Earthquake on Lycanroc will score 300. The decision heuristics are listed below.

- If estimated damage will KO, use it
- Recovery moves are likely if <35% health, less likely between <35%-55%, discouraged if >55%

- Protection moves are scored higher for <30% health to avoid Protect spam
- Set-up moves are likely when health is >70% and unlikely below
- Hazard moves are scored higher when it is not already set and there are 4+ Pokemon on the enemy team left to make it set early
- Hazard removals are likely when hazards are present and even more if team has Flying, Fire, Ice, or Bug Pokemon
- Screens are likely when healthy is >60%
- Status moves check if the opponent is not already statused and scores higher.
- Stat removals like Haze are scored highly if enemy has 2+ stats e.g 1+ in Att and Spe or 2+ in Spe
- Phasing moves are scored higher to force swap boosted enemies
- Priority Finishers are likely if enemy is <25% health
- Pivot moves are likely when <50% health
- Disruption when enemy is not affected always have a boosted score
- Emergency Switch occurs when best move has been used, current health <20% and other Pokemon has >40% health
- Hazard Sensitive Switching scores certain Pokemon less to switch to if specific hazard is set
- Speed Checker for base speed multiplied by any boosts to increase or decrease more viability e.g set-up moves are more likely if faster
- Weather synergy codes moves to use and abilities to switch in for that are boosted in each condition are scored more and penalties for moves that are nerfed e.g switch in Floatzel with Swift Swim and use Waterfall in Rain and enemy team swap out Infernape
- Bad Matchups are considered and switching out is scored higher if a bench Pokemon can resist the enemy STAB better

Results:

This section is the results for the IQV, the initial Champion team performance, the final team performance, a summary table, and the multiple linear regression table on MWE and Stat Total's effect on win rate. Further details such as the 48 test Pokemon, the test teams, initial Champion teams and new Champion teams are in the text files attached with this paper.

Kanto:

Blue - *Blastoise, Pidgeot, Magneon, Exeggutor, Arcanine, and Scyther: MWE 1.11 from 1.08. Stat difference of -20 (3009 from 3029).*

Blue will be played and determined with the Gen 3 games. Gen 1 doesn't have abilities, held items, and natures. Makes things kind of boring. Some conditions for what Blue have to have I considered are Blastoise and Pidgeot are mandatory. Throughout his and Red's rivalry, both have been shown to have Blastoise and Charizard respectively. In Heart Gold/Soul Silver his Ace is a Pidgeot.

Blue's MWE has slightly increased his team's weakness exposure, and his total stat distribution is lower (-20). I hypothesise he will perform marginally worse. Instead, Blue has performed better by almost 2.5%.

Kanto Test Teams vs. Blue Initial	Winrate (Blue)
1	4/2000 / 0.2%
2	2/2000 / 0.1%
3	379/2000 / 19.0%
4	2/2000 / 0.1%
5	4/2000 / 0.2%
6	111/2000 / 5.6%
7	1316/2000 / 65.8%
8	12/2000 / 0.6%
	Total (Mean) = 228.75/2000 - 11.44%

Kanto Test Teams vs. Blue Improved	Winrate (Blue)
1	97/2000 / 4.9%
2	28/2000 / 1.4%
3	72/2000 / 3.6%
4	102/2000 / 5.1%
5	129/2000 / 6.5%
6	428/2000 / 21.3%
7	1216/2000 / 61.0%
8	151/2000 / 7.6%
	Total (Mean) = 277.875/2000 - 13.89%

Johto:

Lance - *Dragonite, Tyranitar, Scizor, Tangrowth, Charizard, Gengar*: MWE of 1.16 from 10.20. Stat difference of -120 (3269 from 3389).

Like the Blue section, I will not be using the original generation for Lance. Instead, I will use Gen 4 Lance in Heart Gold / Soul Silver. If you consult BLourenco's survey in Johto you'll see Snorlax in 32nd and Charizard in 38th. You must leave Johto to obtain them and thus are not included in the test data.

Lance's enormous reduction in MWE should translate into a huge increase in win rate.

However, his team has taken a 120-point difference down as he had three Dragonites with 600 stats each. Lance's win rate increased significantly from 2.8% to 57.14%. His data seem to support MWE and that Stat Total is not as important.

Johto Test Teams vs. Lance Initial	Winrate (Lance)
1	6/2000 / 0.3%
2	19/2000 / 1%
3	8/2000 / 0.4%
4	49/2000 / 2.5%
5	109/2000 / 5.4%
6	256/2000 / 12.8%
7	1/2000 / 0.1%
8	0/2000 / 0%
	Total (Mean) = 56/2000 - 2.8%

Johto Test Teams vs. Lance Improved	Winrate (Lance)
1	810/2000 / 40.5%
2	982/2000 / 49.1%
3	451/2000 / 22.6%
4	1409/2000 / 70.5%
5	1454/2000 / 72.7%
6	1294/2000 / 64.7%

7	1346/2000 / 67.3%
8	1396/2000 / 69.8%
	Total (Mean) = 1142.75/2000 - 57.14%

Hoenn:

Steven - *Metagross, Salamence, Manectric, Wailord, Shiftry, Camerupt*: MWE of 0.58 from 1.20.
Stat difference of +70 (3155 from 3085).

Metagross, being only accessible after beating the Champion, is not included in the test data. Steven's MWE has had a moderate decline, and the Stat Total has increased. My theory expects an increase in win rate. Steven has increased his win rate by over 10%. Less than I expected considering Lance's results.

Hoenn Test Teams vs. Steven Initial	Winrate (Steven)
1	603/2000 / 30.1%
2	435/2000 / 21.6%
3	324/2000 / 16.2%
4	477/2000 / 23.8%
5	255/2000 / 12.8%
6	330/2000 / 16.5%
7	194/2000 / 9.7%
8	1343/2000 / 67.1%
	Total (Mean) = 495.125/2000 - 24.76%

Hoenn Test Teams vs. Steven Improved	Winrate (Steven)
1	160/2000 / 8.0%
2	212/2000 / 10.6%
3	1087/2000 / 54.3%
4	212/2000 / 10.6%
5	1366/2000 / 68.3%

6	313/2000 / 15.7%
7	954/2000 / 47.7%
8	1389/2000 / 69.5%
	Total (Mean) = 711.625/2000 - 35.58%

Wallace - *Milotic, Vileplume, Lunatone, Machamp, Salamence, and Aggron*: MWE of 0.65 from 8.33. Stat difference of +82 (3125 from 3043).

I replaced Wallace's Ludicolo's Double Team with Protect as Showdown does not allow evasion. There is no "gen3anythinggoes" to allow evasion either.

My expectations of Wallace's performance are greater than Steven's due to the much larger change in MWE. Wallace's win rate increase is just over double that of Steven's. This data supports MWE and compared to Steven could show some scale of effect.

Hoenn Test Teams vs. Wallace Initial	Winrate (Wallace)
1	22/2000 / 1.1%
2	207/2000 / 10.3%
3	64/2000 / 3.2%
4	101/2000 / 5.1%
5	1761/2000 / 88.0%
6	521/2000 / 26.1%
7	796/2000 / 39.8%
8	1446/2000 / 72.3%
	Total (Mean) = 614.75/2000 30.74%

Hoenn Test Teams vs. Wallace Improved	Winrate (Wallace)
1	536/2000 / 26.8%
2	1102/2000 / 55.1%
3	446/2000 / 22.3%
4	644/2000 / 32.2%

5	1263/2000 / 63.1%
6	974/2000 / 48.7%
7	1427/2000 / 71.4%
8	1887/2000 / 94.9%
	Total (Mean) = 1034.875/2000 - 51.74%

Sinnoh:

Cynthia - *Garchomp, Spiritomb, Bastiodon, Rotom Wash, Heracross, and Magmortar: MWE of 0.49 from 0.67. Stat difference of -70 (3140 from 3210).*

This team performs marginally worse than her original. I will have to concede that this variable does not hold in this case. Cynthia remains the uncrowned Champion across the games. Completely contradictory evidence compared to Steven and Wallace.

Sinnoh Test Teams vs. Cynthia Initial	Winrate (Cynthia)
1	155/2000 / 7.7%
2	360/2000 / 18.0%
3	105/2000 / 5.2%
4	430/2000 / 21.5%
5	358/2000 / 17.9%
6	1376/2000 / 68.8%
7	1006/2000 / 50.1%
8	1323/2000 / 66.1%
	Total (Mean) = 639.125/2000 - 31.96%

Sinnoh Test Teams vs. Cynthia Improved	Winrate (Cynthia)
1	82/2000 / 4.1%
2	169/2000 / 8.5%
3	32/2000 / 1.6%

4	731/2000 / 36.5%
5	1150/2000 / 57.5%
6	148/2000 / 7.4%
7	958/2000 / 47.9%
8	1403/2000 / 70.2%
	Total (Mean) = 584.125/2000 - 29.21%

Unova:

Ghetsis - *Hydreigon, Carracosta, Ferrothorn, Chandelure, Braviary, and Krookodile*: MWE of 0.61 from 0.43. Stat difference of +76 (3153 from 3087).

The win rate of Ghetsis has improved mildly by just under 6%. This is unexpected as the variable predicted worse performance. Perhaps the Stat Total has affected this.

Unova 1 Test Teams vs. Ghetsis Initial	Winrate (Ghetsis)
1	171/2000 / 8.6%
2	405/2000 / 20.3%
3	190/2000 / 9.5%
4	388/2000 / 19.4%
5	651/2000 / 32.6%
6	354/2000 / 17.7%
7	485/2000 / 24.2%
8	534/2000 / 26.7%
	Total (Mean) = 397.25/2000 - 19.86%

Unova 1 Test Teams vs. Ghetsis Improved	Winrate (Ghetsis)
1	530/2000 / 26.5%
2	487/2000 / 24.3%
3	197/2000 / 9.9%

4	481/2000 / 24.0%
5	674/2000 / 33.7%
6	671/2000 / 33.6%
7	280/2000 / 14.0%
8	816/2000 / 40.8%
	Total (Mean) = 517/2000 - 25.85%

Iris - *Haxorus, Chandelure, Lucario, Swanna, Excadrill, and Hydreigon*: MWE of 0.79 from 1.69. Stat difference of -91 (3166 from 3257).

Iris' win rate almost doubled. The theory holds here but the scale for Stat Total is difficult to tease apart now.

Unova 2 Test Teams vs. Iris Initial	Winrate (Iris)
1	294/2000 / 14.7%
2	227/2000 / 11.4%
3	271/2000 / 13.6%
4	293/2000 / 14.6%
5	1009/2000 / 50.4%
6	549/2000 / 27.5%
7	494/2000 / 24.7%
8	598/2000 / 29.9%
	Total (Mean) = 466.875/2000 - 23.34%

Unova 2 Test Teams vs. Iris Improved	Winrate (Iris)
1	634/2000 / 31.7%
2	454/2000 / 22.7%
3	1128/2000 / 56.4%
4	1284/2000 / 64.2%

5	884/2000 / 44.2%
6	919/2000 / 46.0%
7	866/2000 / 43.3%
8	1287/2000 / 64.3%
	Total (Mean) = 932/2000 - 46.60%

Kalos:

Diantha - *Gardevoir, Drapion, Steelix, Conkledurr, Lanturn, and Staraptor: MWE of 0.45 from 2.87. Stat difference of -176 (2978 from 3154).*

Some mega stones are added after X and Y. They include Beedrillite, Pidgeotite, Salamencite, and Galladite. As I spoke about Snorlax and Charizard against Lance, if it is not available from the region, it will not be included in the training data nor the Champion team.

Though the large improvement of MWE, Diantha's win rate has tanked to less than a third. Her Stat Total decreased a lot as well and might be the reason for this.

Kalos Test Teams vs. Diantha Initial	Winrate (Diantha)
1	112/2000 / 5.6%
2	630/2000 / 31.5%
3	525/2000 / 26.3%
4	959/2000 / 47.9%
5	462/2000 / 23.1%
6	953/2000 / 47.6%
7	410/2000 / 20.5%
8	309/2000 / 15.4%
	Total (Mean) = 545/2000 - 27.25%

Kalos Test Teams vs. Diantha Improved	Winrate (Diantha)
1	3/2000 / 0.2%
2	252/2000 / 12.6%

3	131/2000 / 6.6%
4	152/2000 / 7.6%
5	233/2000 / 11.7%
6	328/2000 / 16.4%
7	109/2000 / 5.4%
8	103/2000 / 5.1%
	Total (Mean) = 163.875/2000 - 8.19%

Alola:

Kukui - *Incineroar, Dhelmise, Skarmory, Raichu-Alola, Goodra, and Whiscash: MWE of 0.48 from 1.72. Stat difference of -42 (3065 from 3107).*

Kukui is paired with an Incineroar in Masters EX so the team will have that. It is also a strong one so it can give his initial team a boost. A bit of a debate that might come up is Silvally. First and only Pokemon to be retroactively stamped as a legendary. Kukui won't be able to have it. But I will have it available for the test teams.

Kukui has supported the MWE variable with a mild increase to his win rate after an improved MWE. His stats have lowered marginally.

Alola Test Teams vs. Kukui Initial	Winrate (Kukui)
1	189/2000 / 9.5%
2	345/2000 / 17.2%
3	106/2000 / 5.3%
4	477/2000 / 23.8%
5	405/2000 / 20.3%
6	41/2000 / 2.1%
7	616/2000 / 30.8%
8	182/2000 / 9.1%
	Total (Mean) = 295.125/2000 - 14.76%

Alola Test Teams vs. Kukui Improved	Winrate (Kukui)
1	195/2000 / 9.8%
2	222/2000 / 11.1%
3	463/2000 / 23.2%
4	297/2000 / 14.8%
5	417/2000 / 20.8%
6	229/2000 / 11.5%
7	1019/2000 / 50.9%
8	509/2000 / 25.5%
	Total (Mean) = 418.875/2000 - 20.94%

Regression:

Below is a table of the changes in MWE, Stat Total, and Win Rate and its Regression Table. Underneath that is a summary of the results from a multiple linear regression with these values.

Champion	MWE Difference	Stat Total Difference	Change in Win Rate
Blue	+0.03	-20	+21.42%
Lance	-9.04	-120	+1940.71%
Steven	-0.62	+70	+43.70%
Wallace	-7.68	+82	+68.31%
Cynthia	-0.18	-70	-8.60%
Ghetsis	+0.18	+76	+30.16%
Iris	-0.90	-91	+99.66%
Diantha	-2.42	-176	-69.94%
Kukui	-1.24	-42	+34.80%

SUMMARY OUTPUT:

Regression Statistics

Multiple R	0.759402
R Square	0.576691
Adjusted R Square	0.435588
Standard Error	4.804557
Observations	9

ANOVA

	df	SS	MS	F	Significance F
Regression	2	188.6877383	94.34383	4.08702	0.075853
Residual	6	138.5026377	23.08377		
Total	8	327.1903			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-1.24429	2.048015	-0.60756	0.565749	-6.25562	3.76702	-6.25562	3.76702
MWE	-1.28104	0.492965	-2.59865	0.040736	-2.48728	-0.0748	0.0748	2.48728

Stat Total	-	0.018447	-	0.407	-0.06158	0.0286	-	0.02869
	0.0164		0.891	095		96	0.06158	6
	4		29					

For MWE, there was no significant relationship to win rate $t(8) = -2.38$, $p = .06$ and Stat Total too had no significant relationship $t(8) = -0.82$, $p = .44$ with the entire model too being insignificant $R^2(2) = .57$, $p = .11$. The MWE p -value doesn't appear too bad. However, Lance's scores have skewed this data heavily as both MWE and Stat Total were even less with MWE, Stat Total, and the whole model producing worse significance $t(8) = -0.009$, $p = .304$, $t(8) = 0.237$, $p = .291$, $R^2(2) = .289$, $p = .506$.

Limitations and Discussions:

A lot of my decisions have exposed this paper to potentially significant flaws.

Local Maximising of Test Data: As mentioned earlier, the best test teams were probably not made for this paper. The computational load to make a Swiss tournament simulation to determine the best test teams was too much for what was not the main focus of the paper. I just need enough teams that are diverse and then to measure a difference in win rate against them. All that mattered was a change in performance of the Champions. It would be interesting to figure out the perfect combination of each 48 selected but I'm too tired of waiting for algorithms and data for the near future.

Zoroark: The teams' orders are coded randomly each battle. This produces a disadvantage of Zoroark's ability Illusion appearing as the last party member when sent out until it is hit. I really couldn't be bothered to set it up with good Illusion Pokemon e.g Fighting types to bait Psychic moves. Regardless it made itself onto both sets of test teams.

Items: It would be interesting to see how much competitive items could change win rates. For the Champion teams, only the Aces have items and they are the same as the base games' choices. It would be interesting to see the increases in win rates when given competitive items.

No Team Preview and Spectating: There were ways to spectate games. However, I opted out of doing this for ease of getting this project done. This has cost me knowing whether every move has been coded for proper use during battles and the typing of Hidden Power being its intended type. However, I'm confident that this wouldn't be a pressure point that could have made MWE work.

Archetypes and Move Analysis: The biggest flaw of this paper is that I have not taken an intentional analysis into the archetypes of the Pokemon used. I did not design the teams with the intent of sweepers, walls, stalls, etc. Probably a thorough investigation into them would yield a better model and explanation of MWE's failings. Moves are collinear with archetypes, but it is the type coverage that is the problem. I just intuitively picked out as many attacks of different

types. These two factors have become a lot more important as I finished collecting the data and further investigation would provide more insight.

Conclusion:

There is no relationship between win rate and MWE or Stat Total. Sucks that I didn't discover some cracked new technique for team building but I had lots of fun making this paper and I am grateful to you for reading it, as incoherent as it might be in some points. Thank you, it means a lot.

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Postscript:

This section is reserved for small hypotheses I began to ask during the making of this. They include (1) the difference between initial champion mono-type vs. pluri-type teams against the training data and (2) the accuracy of results dependent on how many battles against test data (central limit).

Mono-type vs. Pluri-type:

Lance, Steven, Wallace, and Iris are champions with a specialisation. Dragon, Steel, and Water. They do vary in specialisation as Wallace has six Water types, whereas Lance, Steven, and Iris just have three Dragon, Steel, and Dragon respectively. I was curious about how viable it is for a Champion to specialise and go against my MWE. Funny enough to compare their initial win rates, Lance had the worst win rate of 2.8%, Steven with 24.76%, Iris with 23.34%, and Wallace the most specialised had the highest win rate of 30.74%. I won't conclude on anything due to the confounding variables like archetypes not being considered and as much as I tried to avoid it, potential bias in the test data. After getting rid of their specialisations, all the Champions with them saw an improvement in their win rates. The data I have is extremely limited to make any conclusions however it has opened a worthwhile avenue for more investigation.

Central Limit Theorem:

The central limit theorem is the concept that a big enough sampling will produce the results of a normal distribution. It was important in this study to find the right number of battles to accurately determine the win rates of the initial and improved teams. I tested this on Diantha with 10, 100, 1000, and 10000 battles for her test data and measured the variance of each result. I unfortunately did not get a visualisation of the bell curves, but underneath is a table of each test team (y-axis) for Diantha and her win rate (x-axis) for each power of battles done for each team.

	10	100	1000	10000
Team 1	0	4	55	490
Team 2	1	40	339	3281

Team 3	2	18	259	2493
Team 4	3	42	477	4715
Team 5	2	25	220	2200
Team 6	3	47	448	4711
Team 7	1	16	199	2080
Team 8	2	19	165	1534

I determined that to ease computational load as it took five hours for me to run 80000 for one Champion to settle on 2000 battles for each test team. The difference of 1000 to 10000 indicated an acceptable deviation (I could not be bothered to run multiple times to accurately calculate). This was also validated from rerunning the actual algorithm and having very low movement across all test teams and champions.

Credits:

BLourenco's Survey - infogr.am/welcome_your_first_project-84148

iamBGS' Hidden Power Guide - <https://gamefaqs.gamespot.com/boards/696959-pokemon-x/67827372>

hsahovic's Poke-Env - <https://github.com/hsahovic/poke-env>

PokemonDB - <https://pokemondb.net/>

Serebii - <https://www.serebii.net/>

Couldn't have done it without these ^